Task 1: GPS and Navigation Integration

- Simulate GPS latitude and longitude coordinates.
- Generate UAV position information.
- Calculate the difference between the current position and target waypoint.
- Monitor navigation accuracy.

Task 2: Waypoint-Based Autonomous Navigation

- Define multiple waypoints.
- Simulate UAV movement between waypoints.
- Monitor the current and target positions.
- Verify waypoint arrival in simulation.

Task 3: Real-Time Flight Monitoring Dashboard

Create a MATLAB dashboard displaying:

Parameter	Description
Latitude	Current simulated latitude
Longitude	Current simulated longitude
Altitude	UAV height
Speed	UAV velocity
GPS Position	Current location
Waypoint	Target location
Flight Status	Flying / Waypoint Reached

Task 4: Flight Route and Tracking Accuracy

Analyze:

- Actual simulated route.
- Desired waypoint route.
- Position tracking error.
- Distance between waypoints.
- Navigation accuracy.

Task 5: UAV Mission Planning

Use MATLAB to:

- Define a simulated flight area.
- Select starting and destination points.
- Generate waypoint coordinates.
- Simulate the planned mission.
- Visualize the flight path.

Task 6: Flight Performance Analysis

Generate graphs for:

1. Altitude vs Time.
2. Speed vs Time.
3. Latitude vs Longitude.
4. Position Error vs Time.
5. Flight Stability vs Time.

Task 7: Documentation

Prepare:

- System architecture.
- MATLAB source code.
- Simulation screenshots.
- GPS navigation results.
- Flight monitoring dashboard.
- Performance analysis.
- Final project report.

Final Project Outcome

At the end of this project, you will have a MATLAB-based UAV simulation that demonstrates:

- Autonomous waypoint navigation.
- GPS-based position tracking.
- Flight parameter monitoring.

- Flight route visualization.
- Navigation accuracy analysis.
- Flight performance reporting.